



Society of Petroleum Engineers

SPE-227007-MS

Pond Water Based Acid Fracturing: Novel Approach to Improve Sustainability

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

It has been widely recognized on a global scale that both fracturing water-based fluid systems and acid fluid systems predominantly utilize fresh water as their main resource. One of the most promising sustainable solutions is the innovative use of pond water for fracturing treatments, which has shown great potential in reducing reliance on fresh water sources. The concept of utilizing produced water as a base fluid for domestic high-pressure, high-temperature (HPHT) fracturing applications represents a novel advancement in the field. Encouraging results from fluid research, alongside extensive laboratory testing, have demonstrated the viability of substituting fresh water with alternative sources. This led to successful field trials involving pond water-based fracturing operations, aimed specifically at conserving fresh and groundwater resources. Both zirconate crosslinked and borate crosslinked fluids were meticulously reformulated to provide optimal rheological properties when mixed with pond water. The trial campaign not only proved the reliability of these systems at elevated bottom hole temperatures but also demonstrated their compatibility with acid fluid systems, passing all necessary tests without significant alterations. Rheology data and compatibility testing have convincingly shown the effectiveness of the optimized formulation for pond water in high-temperature wells. Several trial stages were executed in rig-less stimulation operations, remarkably without any incidents related to health, safety, or environmental concerns. The introduction of pond-sourced water into fracturing operations at high temperatures (reaching up to 315°F) in vertical wells was successfully implemented within an acid fracturing fluid system. The first trial well was subsequently opened for flowback, yielding promising results and an excellent gas production rate. Overall, the pond water trial campaign significantly lowered the consumption of fresh groundwater, achieving impressive average savings of around 1,700 barrels of fresh water for each fracturing treatment conducted. The authors estimate a potential conversion rate of up to 18% of conventional fracturing operations during the initial commercial year, with further expansion of the method anticipated. Ultimately, the estimated potential for saving fresh water could reach as high as 50% annually. This paper presents a comprehensive overview of the novel methodology employed in utilizing pond water as a source for acid fracturing fluid systems applied in HPHT carbonate formations. This innovative approach contributes significantly to sustainable water conservation efforts, addressing the

excessive use of fresh water by leveraging produced water (Ajayi et al., 2013; Economides & Nolte, 2000; Kalgaonkar et al., 2021).

Introduction

Traditional fracturing fluid systems rely heavily on fresh water, leading to serious concerns regarding groundwater depletion and the long-term availability of this vital resource. Recent studies, such as SPE-219031-MS (Rajendra, 2024), emphasize the critical importance of exploring alternative water sources to support sustainability and promote a circular water economy within the hydraulic fracturing sector. This paper introduces a groundbreaking approach that leverages pond water as a sustainable alternative for acid fracturing fluid systems, specifically targeting high-pressure, high-temperature (HPHT) applications. Hydraulic fracturing has become the preferred stimulation technique for enhancing productivity in tight gas formations, often necessitating the use of several thousand barrels of freshwater per frac stage (Economides & Martin, 2007). To mitigate the potential burden on freshwater resources, the oil and gas industry has increasingly turned its attention to alternative water sources, including pond water sourced from natural gas liquids (NGL) plants. Although alternative water sources are abundant, their usability in hydraulic fracturing treatments has historically been limited due to compatibility concerns with fracturing fluid additives and reservoir fluids. This paper aims to address these concerns by presenting a compelling case for the application of pond water in hydraulic fracturing processes.

Production of Pond Water

The production of pond water in the oil and gas industry starts with the collection of produced water, a byproduct of hydrocarbon extraction that often contains dissolved salts, hydrocarbons, and contaminants. This produced water is directed into large, shallow evaporation ponds, where it is spread out to enhance solar evaporation. This method significantly reduces the volume of liquid waste requiring treatment or disposal, particularly in regions with high evaporation rates (Punder & Veil, 2007). Additionally, seawater is increasingly considered as an alternative water source for operations, including enhanced oil recovery (EOR), helping to reduce pressure on freshwater resources. However, challenges such as salinity management and potential impacts on reservoir performance must be addressed. The industry is under increasing regulatory pressure to manage water resources sustainably, complying with environmental regulations regarding the discharge of produced water and the use of alternative sources (US Energy Information Administration, 2023). Research, including studies by Kalgaonkar et al. (2023), focuses on the efficiency and environmental implications of these practices.

Filtration

The pond water is first passed through a series of filtration units designed to remove large particulates, such as sand, silt, and debris. This step is critical to prevent clogging or damage to equipment during the fracturing process.

Chemical Treatment. After filtration, the water may undergo chemical treatment to remove specific contaminants. This process can include adding flocculants or coagulants to help eliminate dissolved solids and improve water clarity. Additionally, adjustments to pH levels may be made to enhance the effectiveness of fracturing fluid additives.

Compatibility Testing

Before the water is deemed suitable for use in fracturing fluids, it undergoes compatibility testing with various chemical additives typically used in fracturing operations. This ensures that the pond water will not adversely react with these additives, which could compromise the performance of the fracturing fluid (Table. 4).

Quality Control

Comprehensive quality control measures are implemented throughout the processing steps. Samples of the treated pond water are analyzed for various parameters, including total dissolved solids (TDS), salinity, and concentrations of specific contaminants. This ensures that the water meets the stringent quality standards required for effective and safe hydraulic fracturing. (Table. 3)

Operational overview /Transportation to Location

After processing and treatment, the pond water is transported to the fracturing site using specialized equipment. The transportation process involves several critical steps:

Storage and Loading

Treated Pond water is stored in temporary holding tanks equipped with monitoring systems to track water quality. Before transportation, the water is loaded into tank trucks or specialized water transport vehicles designed to handle fluids with varying viscosities.

Pipeline Systems

In cases where large volumes of water are required, a network of pipelines may be employed to transport the pond water directly from the treatment facility to the well site. These pipelines are constructed from durable materials to withstand the pressures and chemical properties of the pond water.

Monitoring During Transport

Throughout the transportation process, real-time monitoring systems are utilized to track the quality and flow of the pond water. Parameters such as pressure, temperature, and flow rates are continuously monitored to ensure that the water remains within acceptable limits and is delivered safely and efficiently.

Final Quality Check

Upon arrival at the fracturing location, a final quality check is performed to ensure that the pond water meets all required specifications before it is mixed with chemical additives and proppants for the fracturing job. This step is crucial to confirm that the water retains its integrity and is ready for efficient use in the fracturing process (Tables. 3&6).

The rigging up of equipment and water collection process for pond water (Fig. 1) begins with the C-Pump 5x6 (Fig.2), which is set up with two 4-inch, 20-foot hoses. Two operators, consisting of one operator and one helper, are assigned for the day shift to operate the C-Pump. Essential tools required include thread, wing connections, a hammer, and a wire brush, along with four belts to secure the C-Pump to the pipe and prevent any fall into the pit due to vibration. A fire extinguisher and diesel for refueling the C-Pump are also necessary. The C-Pump is connected to a 4-inch flange (Fig.3), and a forklift is needed for assistance. Upon arrival at the pond water source, the C-Pump and hoses are offloaded, with hoses connected from the pond water pipe outlet to the C-Pump and then to the water transport. The C-Pump operates at a reduced rate of 1000 RPM, facilitating smooth water transfer. It takes approximately 35 minutes to fill a 32,000-liter transport (Fig. 4).



Figure 1—Pond Water



Figure 2—C-Pump



Figure 3—Flange and 4" Weco wing



Figure 4—Water transport (32,000 liters)

Operational Efficiency

The introduction of pond water into fracturing operations carries significant implications for the conservation of freshwater resources. On average, each fracturing treatment successfully saves approximately 1,700 barrels of fresh water, representing a substantial achievement in the pursuit of sustainability within the industry. Notably, the use of pond water utilizes the same typical fracturing equipment designed for fresh water, eliminating the need for any special equipment. This streamlines operations and reduces costs associated with transitioning to alternative water sources. Additionally, the operation employs specialized equipment designed specifically for high-temperature, high-pressure environments, ensuring optimal performance throughout the fracturing process.

Design phase

In preparation for the candidate well to be treated with pond water a simulation software was used to design the treatment as mentioned by the study, [Nurlybayev et al. \(2023\)](#) explore innovative solutions to the challenges faced in hydraulic fracturing within high-pressure, high-temperature (HPHT) wells. The authors present a comprehensive analysis of a newly developed fracturing fluid system tailored to enhance performance in these demanding environments. By utilizing a full 3D simulator in their methodology, they are able to evaluate different fracturing designs and their effectiveness in real-world applications. Their findings demonstrate that the implementation of advanced fluid systems significantly improves the efficiency of fracturing treatments, paving the way for more sustainable practices in the oil and gas industry.

Results/ Production

Field trials conducted during this study demonstrated that the pond water-based fracturing systems-maintained reliability at bottom hole temperatures reaching up to 300°F. The compatibility tests revealed that the optimized formulation performed effectively under these challenging conditions. Importantly, the stimulation fluid exhibited excellent post-stimulation flowback performance, indicating the development of a non-damaging frac fluid system derived from evaporation pond water. Additionally, zirconate crosslinker fluid combined with CMHPG is specifically designed for HPHT applications, particularly when bottom hole temperatures exceed 300°F. Crosslink fluid formulations are depicted in [Table 5](#). This fluid system is tailored to meet client specifications and is rigorously tested and adjusted based on field conditions, stability requirements, and temperature variations (see [Fig. 03](#)). Surface field conditions are crucial, especially with temperature fluctuations ranging from 60°F in winter to 110°F in summer. The fluid is optimized for both surface and bottom hole conditions. In winter, as surface temperatures drop to 60°F, it becomes essential to optimize the delaying agent to achieve the necessary crosslinking time within the specified pumping duration while ensuring that bottom hole temperature conditions are also satisfied.

Injection Test Stage-1

Initial WH pressure was 1,400 psi. Injection test started by pumping treated water at 1.0 bpm and then rate was increased to max 15.4.0 bpm. Total volume of treated pond water pumped was 100.0 bbl, Max STP was 9,425 psi and Max BHTP was 14,900 psi.

Main FRAC Stage-1

Initial WH pressure was 1,170 psi. The Pond Water job was performed for Stage-1 (Fig. 6) carbonate formation starting with rate of 10.0 bpm for establishing the rate surface pressure stabilization. After initial acid volumes reached formation pumping rate was increased to max 43.0 bpm as shown for all steps on the treatment (Table. 1). Total fluid pumped was 1,536 bbl as shown on (Table.2). Acid was flushed and overflushed with 321 bbl of Treated Pond water. Max STP was 11,730 psi while the max BHTP was 14,665 psi.

Table 1—Main Treatment schedule

STG #	STAGE NAME	FLUID TYPE	PUMP RATE (BBL/min)	SLURRY VOL (GAL)		TIME (MIN:SEC)
				STAGE	CUM.	STAGE time
1	Establish injection	Treated Pond Water	8.7	924	924	2.5
2	Spearhead acid	28% HCL (275F)	10.1	2,016	2940	4.8
3	Acid Displacement	Treated Pond Water	15.5	4,200	7140	6.5
4	Pad-1	Crosslinked gel 40#	18.8	3,486	10626	4.4
5	Emulsified Acid-1	Emulsified Acid	26.7	2,100	12726	1.9
6	Straight acid-1	28% HCL (275F)	30	2,562	15288	2.0
7	Diversion -1	Diverter	30	1,764	17052	1.4
8	Spacer Water-1	Spacer Water	30	1,050	18102	0.8
9	Pad-2	Crosslinked gel 40#	34.3	3,612	21714	2.5
10	Emulsified Acid-2	Emulsified Acid	39	2,100	23814	1.3
11	Straight acid-2	28% HCL (275F)	42.2	2,646	26460	1.5
12	Diversion -2	Diverter	43	1,764	28224	1.0
13	Spacer Water-2	Spacer Water	43	1,050	29274	0.6
14	Pad-3	Crosslinked gel 40#	41.1	3,612	32886	2.1
15	Emulsified Acid-3	Emulsified Acid	41.7	2,100	34986	1.2
16	Straight acid-3	28% HCL (275F)	43	2,646	37632	1.5
17	Diversion -3	Diverter	43	1,764	39396	1.0
18	Spacer Water-3	Spacer Water	43	1,050	40446	0.6
19	Pad-4	Crosslinked gel 40#	43	3,696	44142	2.0
20	Emulsified Acid-4	Emulsified Acid	43	2,100	46242	1.2
21	Straight acid-4	28% HCL (275F)	43	2,730	48972	1.5
22	Sweep	Crosslinked gel 40#	43	2,058	51030	1.1
23	75% overflush	Treated Pond Water	43	4,494	55524	2.5
24	Flush	Treated Pond Water	24.7	8,988	64512	8.7

Table 2—total fluids pumped

#	Fluid Type	Total Volume	Units
1	Crosslinked gel 40#	16,464.00	gal
2	28% HCL (275F)	12,600.00	gal
3	Emulsified Acid	8,400.00	gal
4	Diverter	5,292.00	gal
5	Spacer Water	3,150.00	gal
6	Treated Pond Water	18,606.00	gal

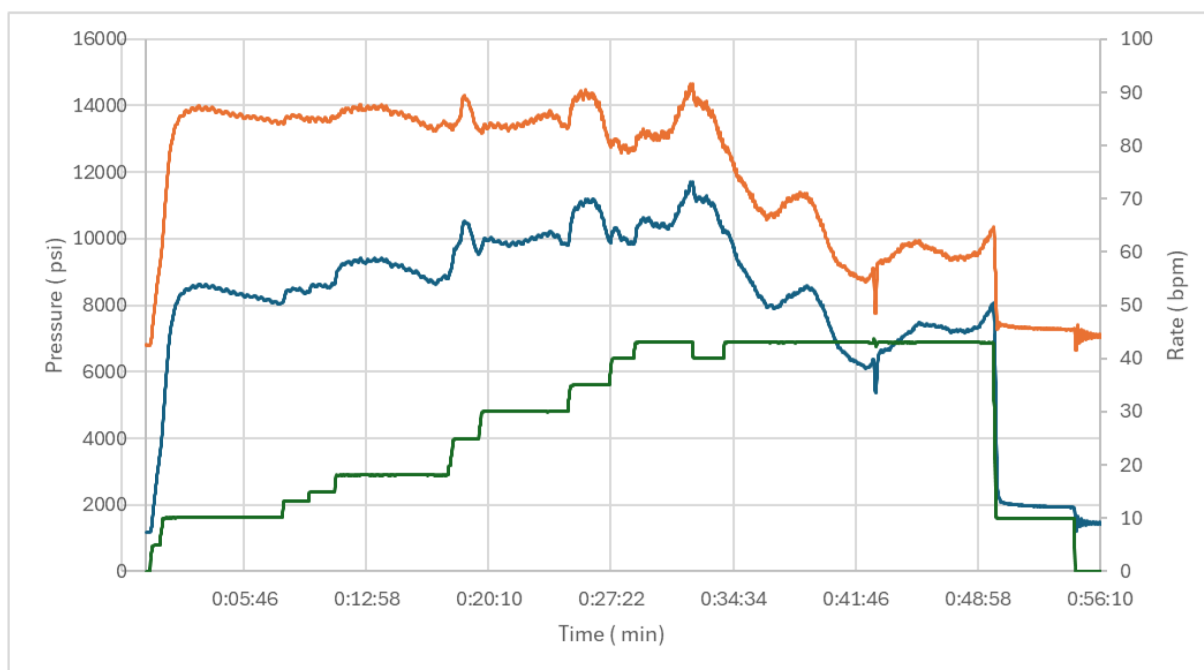


Figure 5—Chart Main Treatment of Stage 1

Table 3—Water analysis

Parameters	Limit	22 nd July sample	1 st August sample	18 th October sample	UoM
Fluid Test Temperature (degF)	35-100	74	75	77.3	deg F
Color (unfiltered sample)	Clear	Clear	Clear	Clear	
pH	6.0-8.0	7.65	8.5	7.84	pH
Specific Gravity	0.99-1.01	1.01	1.03	1.01	g/cm ³
Total Dissolve Solid	<1500	5872	6324	5932	mg/l
Sulphate, SO ₄ ²⁻	<500	1703	1654	1683	mg/l
Chloride, Cl ⁻	<1,200	1697	1835	1790	mg/l
Bicarbonate, HCO ₃ ³⁻	<500	3123	2965	2938	mg/l
Total Hardness	<500	3229	2980	3067	mg/l
Calcium, Ca ²⁺	<250	1783	1620	1679	mg/l
Magnesium, Mg ²⁺	<250	1446	1360	1388	mg/l
Iron, Total	<10	0.09	0.12	1.1	mg/l
Barium, Ba ²⁺	<100	11	14	12	mg/l

Table 4—Compatibility testing for 4 hours

Fluid	Temperature °F	Precipitation or Separation	Compatible
Emulsified Acid	25	No	Yes
	210	No	Yes
Straight Acid	25	No	Yes
	210	No	Yes
Visco-elastic Acid	25	No	Yes
	210	No	Yes

Table 5—Crosslink formulation

Additive	Description	Medallion HT 4500 AT 315F Conc.	Medallion HT 4500 AT 300 Conc.	Medallion HT 4000 AT 300F Conc.	Unit
Water	Pond Water	970	970	970	gpt
Xcide 575	Biocide	0.05	0.05	0.05	gpt
MC MXI 2-3440	Scale Inhibitor	0.1	0.1	0.1	gpt
XLFC-3B	Gelling agent slurry	11.25	11.25	10	gpt
Stim Max C	Fluid Recovery	1	1	1	gpt
Floback Prime	Flowback surfactant	2	2	2	gpt
GS-1L	HT stabilizer	12	11	13	gpt
BF-9L	Buffer	4.1	4.1	3.9	gpt
XLW-60	Crosslinker	1.3	1.3	1.15	gpt
GBW-25	Breaker	0.3	0.5	0.3	gpt

Table 6—Stage-1 Field quality assurance quality control results

Temp (°F)	88.8	87.8								
Color (unfiltered sample)	Clear	Clear								
pH	7.69	7.65								
SG	1.00	1.00								
TDS	6326	5835								
SO ₄ ²⁻ (mg/L)	1650	1672								
Cl (mg/L)	1820	1846								
HCO ₃ ⁻ (mg/L)	2940	2884								
Total Hardness (mg/L)	3070	3140								
Ca ²⁺ (mg/L)	1680	1682								
Mg ²⁺ (mg/L)	1390	1458								
Fe ^{2+,3+} (mg/L)	1.06	0.98								
Ba ²⁺ (mg/L)	12	16								
Part 2: Laboratory Linear Gel QAQC Before Job										
Name of product system mixed	Linear Gel	Linear Gel								
Sampled Date	21-Nov-22	21-Nov-22								
Sampling time	8:05	8:10								
Gellant loading (gal/Mgal)	40	40								
Fluid sampling location	Frac Tank	Frac Tank								
Sample temperature, °F	76.7	77.2								
Fann reading @ 300 rpm	3 min									
	10 min	40	41							
	15 min									
Linear Gel pH	7.56	7.89								
Buffered pH	7.92									
Buffer added (gpt)										
X-link Vortex Closure, min:sec										
X-Link Hangup, min:sec										
X-Link pH										
Part 3: Acid Tank QAQC Before Mixing										
Acid Tank Number	001	002	003	004					Acid Truck	Vol (gal)
Date Test	20-Nov-22	20-Nov-22	20-Nov-22	20-Nov-22					4.0	20,000
Temp (°F)	89	91	90	93						
Titration NaOH	10.00	9.80	10.00	10.00						
Acid Strength (%)	31.7	31.1	31.7	31.7						
Iron Content (mg/L)	5.0	3.5	2.5	9.5						
Part 4: Batch Mixed Fluid QAQC Before Pumping										
Fluid System	28% HCl	28% EA	20% VES	Linear Gel 40						
Frac Tank Number	7	5	6	2						
Volume (bbl)	300	200	130	435						
Titration NaOH	8.50		6							
Acid Strength	27.21		19.72							
Fe ^{2+,3+} (mg/L)	26		10							
Conductivity (S/cm)		0.05								
Linear Gel pH				7.76						
Temp (deg°F)				76						
Gel Loading				40						
Viscosity (cP)				41						

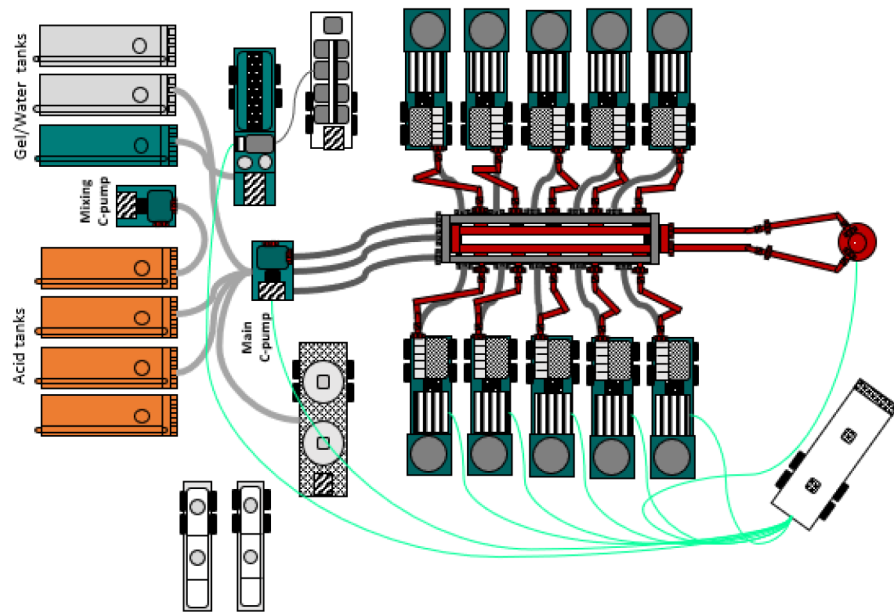


Figure 6—Typical Acid Frac Layout

Well flow back results after the treatment

The well was opened for flowback after a shut in pf 24 hours with a flowback period of 5 days. The well exhibited excellent flowback without showing any formation damage due to the pondwater based fracturing fluid.

critical equipment for acid frac:

Pumping Units: Designed to operate at 35-40 bbl/min with a pressure capacity of 14,500 psi,

- 3 × 3" Discharge Iron: Three lines to be connected to the wellhead.
- 7x 2,000 HHP Frac Pumps: Including backup pumps.
- 1 × Batch Mixer: For mixing fluids.
- 1 × LAS Liquid Additive System: For managing liquid additives.
- 1 × DATAVAN: Equipped with a field laboratory.
- 8 × 500 bbl Frac Tanks: For storing acid and water.
- 1 × HP Manifold: To manage high-pressure operations.
- 2 × Relief Valves: For 2" high-pressure lines.
- 1 × Generator: To provide power.
- 1 × Air Compressor: For pneumatic operations.
- 1 × Treesaver: For well safety.

Typical Acid frac equipment Layout

The typical acid frac layout similar to that used in the pondwater fracturing fluid application is depicted in [Fig. 6](#)

Conclusion

This study presents a robust and sustainable methodology for utilizing pond water in acid fracturing fluid systems, significantly contributing to the reduction of reliance on fresh water in high-pressure, high-temperature (HPHT) carbonate formations. By adopting this approach, the industry can significantly mitigate its environmental footprint, promoting the conservation of precious freshwater resources. Furthermore, this methodology emphasizes that any water source can be transferred, treated, and effectively utilized within the fracturing fluid systems. This flexibility showcases the capability of the industry to handle multiple water resources, including brackish water or produced water, thus enhancing the overall sustainability of hydraulic fracturing operations. The successful integration of typical equipment and innovative operational strategies not only demonstrates the feasibility and effectiveness of this method but also highlights its potential to enhance environmental sustainability. Promising results from the trial campaigns indicate a viable path toward enhancing sustainability in hydraulic fracturing operations, paving the way for wider applications and further research in the field. This methodology not only addresses water scarcity issues but also aligns with environmental stewardship goals, fostering a more sustainable future for energy production and resource management. By prioritizing the use of alternative water sources, the industry can contribute to significant reductions in environmental impact and promote a more sustainable approach to energy extraction. In conclusion, the ability to utilize various water sources not only aids in resource management but also opens up new avenues for research and innovation, ultimately leading to more sustainable practices in the energy sector.

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